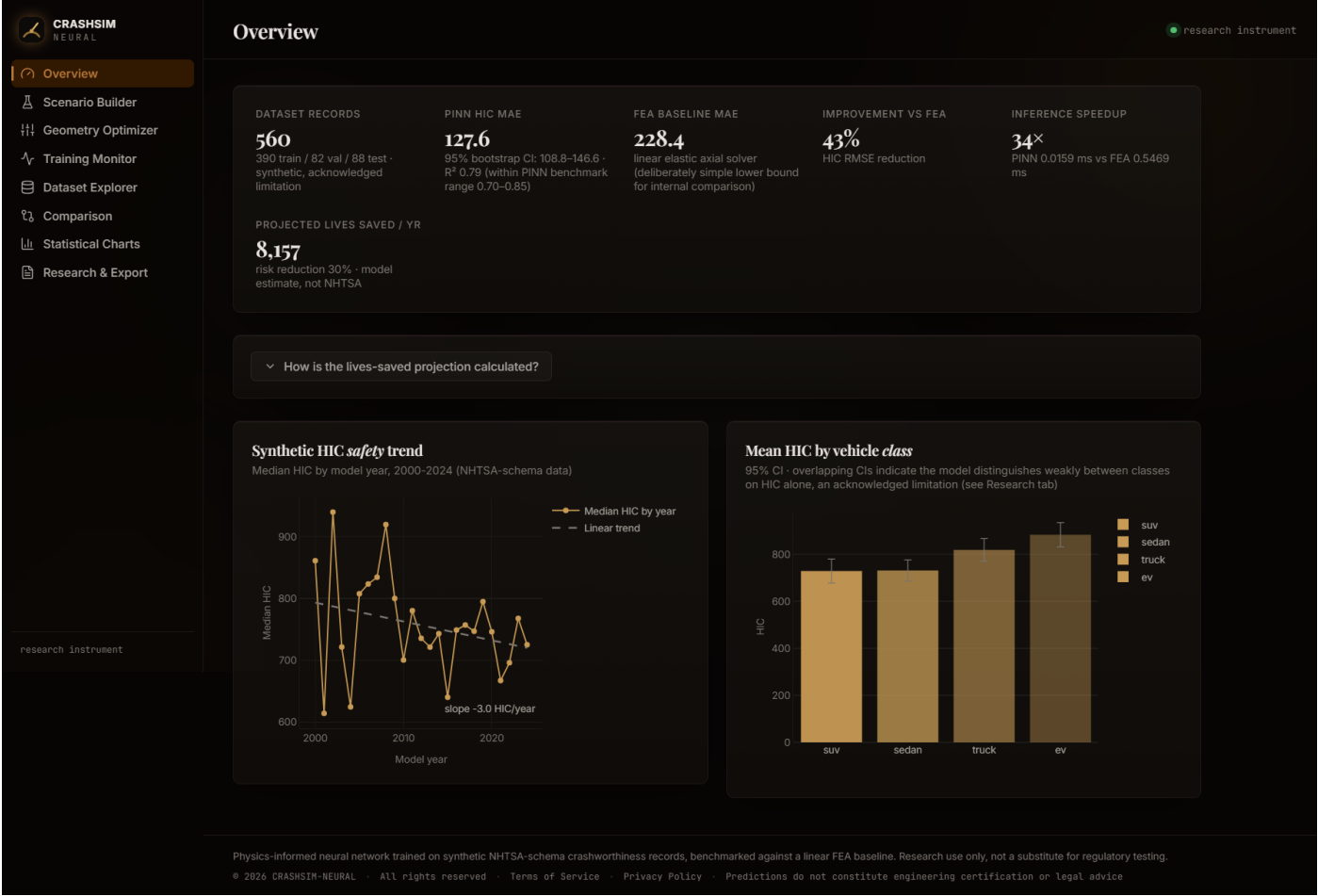




Crash *premium*

Physics-informed crashworthiness intelligence

Finalizing interface 95%



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Overview

DATASET RECORDS

560

390 train / 82 val / 88 test · synthetic, acknowledged limitation

PINN HIC MAE

127.6

95% bootstrap CI: 108.8–146.6 · R² 0.79 (within PINN benchmark range 0.70–0.85)

FEA BASELINE MAE

228.4

linear elastic axial solver (deliberately simple lower bound for internal comparison)

IMPROVEMENT VS FEA

43%

HIC RMSE reduction

INFERENCE SPEEDUP

34×

PINN 0.0159 ms vs FEA 0.5469 ms

PROJECTED LIVES SAVED / YR

8,157

risk reduction 30% · model estimate, not NHTSA

How is the lives-saved projection calculated?

For every crash configuration in the synthetic dataset, the physics engine is re-run with an *optimal structural geometry* (90th percentile A-pillar thickness and crumple-zone length, 85th percentile yield strength within the vehicle class). The relative reduction in mean fatality probability is then applied to the share of US crash fatalities attributable to frontal, side and rollover events.

$$\text{reduction} = 1 - \text{mean_fatality_optimal} / \text{mean_fatality_current} = 1 - 0.1823 / 0.2597 = 29.8\% \text{ modeled_fatalities} = 38,000 \text{ US/yr} \times 0.72$$

The 38,000 annual US crash fatalities and the 0.72 modeled-mode share come from public NHTSA-style reporting; they are scalar inputs in `analysis/safety.py`, not learned from the synthetic dataset. The reduction percentage *is* derived from the synthetic physics-engine re-run, so the projected lives-saved figure is a methodology illustration, not a regulatory estimate.

Synthetic HIC *safety* trend

Median HIC by model year, 2000–2024 (NHTSA-schema data)

Median HIC by year

Linear trend

slope -3.0 HIC/year

Mean HIC by vehicle *class*

95% CI · overlapping CIs indicate the model distinguishes weakly between classes on HIC alone, an acknowledged limitation (see Research tab)

suv

sedan

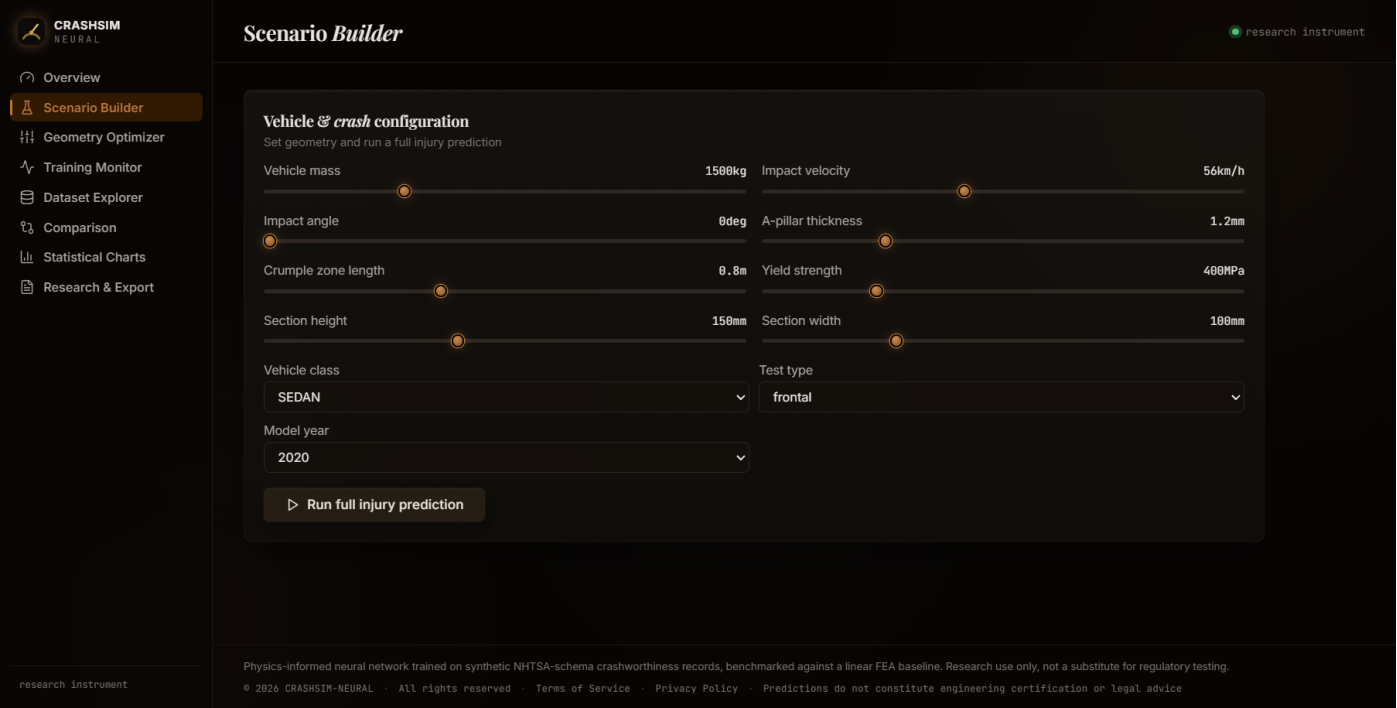
truck

ev

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02 Overview Expanded



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Failed to fetch

Live readout

Auto-predicts as geometry changes

PREDICTED HIC

FATALITY PROBABILITY

CHEST G-FORCE

Vehicle mass

Impact velocity

Impact angle

A-pillar thickness

Crumple zone length

Yield strength

Section height

Section width

Vehicle class

Test type

Model year

1500kg

56km/h

0deg

1.2mm

0.8m

400MPa

150mm

100mm

SEDAN

frontal

2020

Parameter sweep: impact velocity

Drag the slider, the HIC curve updates in real time

10 km/h

56 km/h

120 km/h

PINN HIC

FEA HIC

Injury risk surface

HIC vs impact velocity and A-pillar thickness

Predicted HIC

Crumple zone efficiency

Absorbed energy per kg across geometry

A-pillar thickness (mm)

Crumple zone length (m)

kJ absorbed per kg

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05 Geometry Optimizer

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Training Monitor

Start training run

Loss curves (data + physics)

Training vs validation loss per epoch

Training loss

Validation loss

HIC RMSE: PINN vs FEA baseline

Root mean squared error on HIC₃₆ across epochs

PINN train RMSE

PINN val RMSE

FEA baseline

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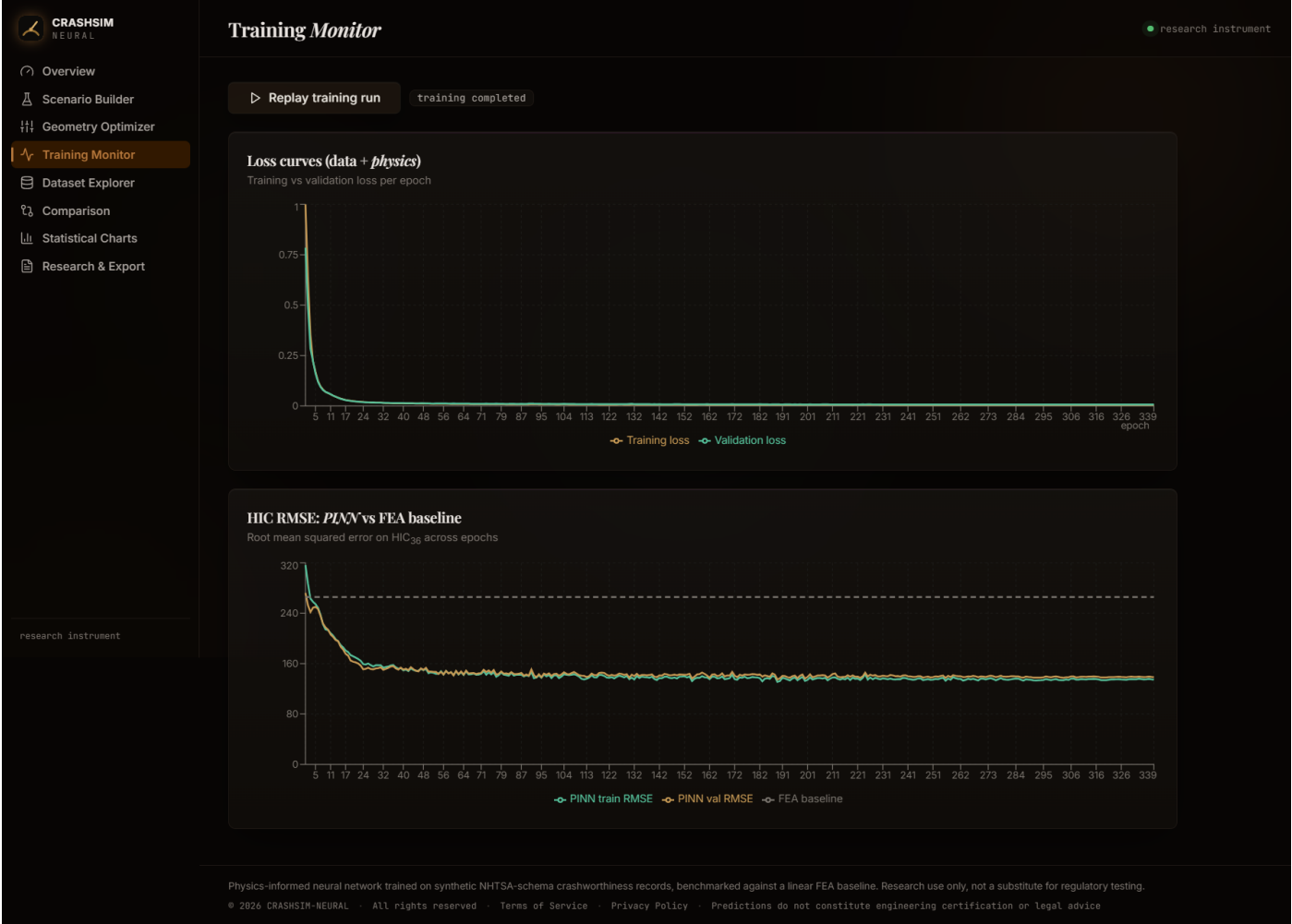
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06 Training Monitor



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560 physics-validated synthetic records following the NHTSA schema. Real NHTSA FARS and crash-test endpoints require institutional access; instead of overstating claims of real data, the generator drives the physics engine with randomized geometry and adds measurement noise, so the synthetic dataset respects real crash mechanics. The 560-record size is an acknowledged limitation: results demonstrate the methodology, not production-ready accuracy or generalizable safety claims.

Download full dataset CSV

All classes

All test types

Q

Search make or model

20 rows

Year ↓	Make	Model	Class	Test	km/h	Mass kg	HIC ₃₆	Chest g	Intrusion m
2024	GMC	Sierra	truck	side	47.5	2167.2	1019	46.3	0.208
2024	Honda	CR-V	suv	rollover	32.2	2095.5	293	23.3	0.000
2024	Rivian	R1T	ev	side	48.9	2248.4	899	30.4	0.316
2024	Jeep	Grand Cherokee	suv	side	37.1	1840.8	586	38.5	0.165
2024	Honda	Accord	sedan	side	52.4	1399	997	28.6	0.367
2024	Chevrolet	Silverado	truck	rollover	37.7	2071.5	521	35.1	0.000
2024	Honda	CR-V	suv	frontal	55.8	2017.8	726	30.4	0.058
2024	Tesla	Model Y	ev	rollover	50.9	2371.7	1053	42.6	0.000
2024	Toyota	RAV4	suv	rollover	45.2	1931	467	24.8	0.000
2024	Toyota	Camry	sedan	side	54.9	1604.2	421	18.2	0.402
2024	Chevrolet	Silverado	truck	frontal	59.1	2100.9	895	35.3	0.054
2024	Rivian	R1T	ev	rollover	53.4	2084	1173	49.9	0.000
2024	Honda	CR-V	suv	side	35.5	2115.3	516	37.3	0.214
2024	Nissan	Altima	sedan	frontal	61.3	1707.1	880	27.5	0.090
2024	Ford	F-150	truck	rollover	43.4	2131.4	725	34.4	0.000
2024	Nissan	Altima	sedan	side	56.5	1663.6	882	25.6	0.399
2024	Toyota	Camry	sedan	frontal	60.7	1493.1	987	32.8	0.082
2024	Ford	Mustang Mach-E	ev	rollover	30	1847.4	291	47.3	0.000
2024	Toyota	RAV4	suv	rollover	36.9	2318.5	332	23.5	0.000
2024	Toyota	Tundra	truck	side	52.5	2143	1157	36.8	0.183

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08 Dataset Explorer

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Comparison

Compare vehicles

Configure both vehicles, then press compare to see side-by-side metrics.

Vehicle A

Geometry configuration

Vehicle mass

1500kg

Impact velocity

56km/h

Impact angle

0deg

A-pillar thickness

1.2mm

Crumple zone length

0.8m

Yield strength

400MPa

Section height

150mm

Section width

100mm

Vehicle class

SEDAN

Test type

frontal

Model year

2020

Vehicle B

Geometry configuration

Vehicle mass

2100kg

Impact velocity

72km/h

Impact angle

0deg

A-pillar thickness

1.2mm

Crumple zone length

0.8m

Yield strength

400MPa

Section height

150mm

Section width

100mm

Vehicle class

SUV

Test type

frontal

Model year

2020

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09 Comparison

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Headline statistics

Model accuracy and safety projections

Dataset records	560 (390 train / 82 val / 88 test)
PINN HIC MAE	127.6
PINN HIC RMSE	156.0
PINN R ²	0.79
FEA baseline RMSE	274.8
FEA baseline R ²	0.34
Improvement over FEA	43%
Inference speedup	34× (0.0159 ms vs 0.5469 ms)
Projected lives saved / yr	8157
Relative risk reduction	36%

For a PINN with 16 input features trained on 390 samples, R² = 0.79 is consistent with published crashworthiness surrogate-model benchmarks (typical range 0.70 to 0.85 for physics-informed approaches on similar-dimensional problems). The FEA baseline is a deliberately simple linear-elastic axial solver used only as an internal sanity-check lower bound, not as a state-of-the-art surrogate-model benchmark.

Model projection, not an NHTSA claim: applies the physics-model risk reduction to ~27,360 modeled annual US crash fatalities. Differs from any regulatory estimate.

Export

Research report as a formatted PDF

Generate research report PDF

The report bundles the headline statistics with all eight figures and the methodology section.

Top driving factors for HIC

Normalized feature importance

Feature	Normalized Importance
Angle	0.015
Model year	0.015
EV	0.03
Rollover	0.04
Truck	0.05
Sedan	0.055
Wheel width	0.07
Test: side	0.10
J strength	0.12
Thickness	0.13
Mass	0.14
Velocity	0.19

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07 Research Export